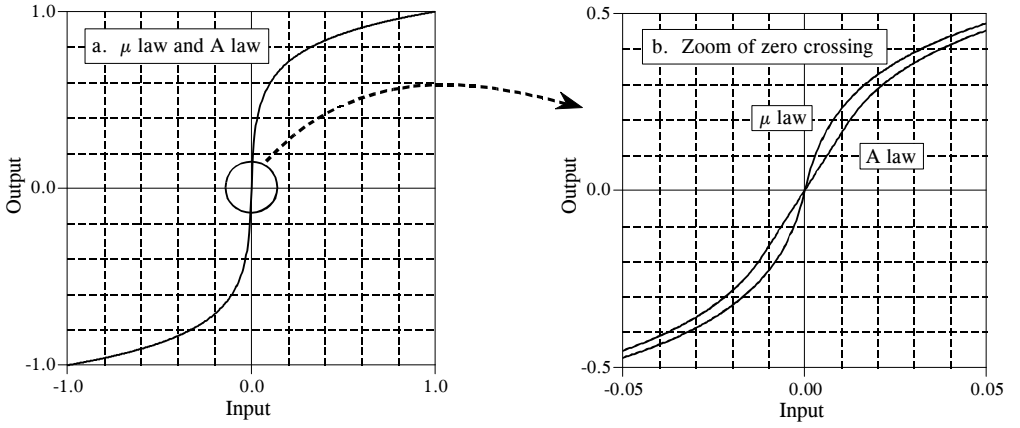

FIGURE 22-7
Companding curves. The μ255 law and "A" law companding curves are nearly identical, differing only near the origin. Companding increases the amplitude when the signal is small, and decreases it when it is large.

voltage across a *pn* diode junction, and then add circuitry to correct for the ghastly temperature drift. Most companding circuits take another strategy: approximate the nonlinearity with a group of straight lines. A typical scheme is to approximate the logarithmic curve with a group of 16 straight segments, called **cords**. The first bit of the 8 bit output indicates if the input is positive or negative. The next three bits identify which of the 8 positive or 8 negative cords is used. The last four bits break each cord into 16 equally spaced increments. As with most integrated circuits, companding chips have sophisticated and proprietary internal designs. Rather than worrying about what goes on inside of the chip, pay the most attention to the pinout and the specification sheet.

Speech Synthesis and Recognition

Computer generation and recognition of speech are formidable problems; many approaches have been tried, with only mild success. This is an active area of DSP research, and will undoubtedly remain so for many years to come. You will be very disappointed if you are expecting this section to describe how to build speech synthesis and recognition circuits. Only a brief introduction to the typical approaches can be presented here. Before starting, it should be pointed out that most commercial products that produce human sounding speech do not *synthesize* it, but merely play back a digitally recorded segment from a human speaker. This approach has great sound quality, but it is limited to the prerecorded words and phrases.

Nearly all techniques for speech synthesis and recognition are based on the model of human speech production shown in Fig. 22-8. Most human speech sounds can be classified as either **voiced** or **fricative**. Voiced sounds occur when air is forced from the lungs, through the vocal cords, and out of the mouth and/or nose. The vocal cords are two thin flaps of tissue stretched across